Austo Automobile
Business Report

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About the Data

1) Data Description

The dataset consists of 1,581 customer records and 14 variables, with each record representing an individual customer.

Variables in the dataset:

Variable	**Description**
Age	Age of individual in years
Gender	Gender of individual (Male or Female)
Profession	Occupation of the individual
Marital_status	Marital status of individual (Married or Single)
Education	Qualification of individual (Graduate or Post Graduate)
No_of_Dependents	The number of dependents that the individual supports financially
Personal_loan	Indicates whether the individual has a personal loan
House_loan	Indicates whether the individual has a House loan
Partner_working	Indicates if the individual's partner is employed
Salary	Individual's salary or income
Partner_salary	Salary of the individual's partner (if applicable)
Total_salary	The combined salary of the individual and their partner (if applicable)
Price	The price of a product or service
Make	Type of automobile

Table 1 Variables in the dataset

2) Data Types

The dataset contains both categorical and numerical variables.

Categorical Variables:
Gender, Profession, Martial_status, Education, Personal_loan, House_loan, Partner_working, Make

Numerical Variables:
Age, No_of_Dependents, Salary, Partner_salary, Total_salary, Price

Target Variable:
The primary or target variable for this analysis is – Make – the type of automobile (SUV, Hatchback, Sedan)

The remaining variables are used to understand the customer characteristics and patterns related to automobile preference.

3) Data Preparation

The dataset was imported into a Python environment and reviewed to understand its structure and characteristics before any analysis was conducted.

It was loaded into a Pandas DataFrame to enable data manipulation and analysis; standard libraries such as Pandas, Matplotlib, and Seaborn were imported to support data exploration and visualization.

Dataset Overview

Initial inspection of the dataset was conducted using the functions as follows:

- head( ) – to view the first few records to ensure the right dataset had been imported
- shape – to determine the number of rows and columns in the dataset
- info( ) – to determine data types and identify missing values
- duplicated( ) – to check if there are any duplicate entries in the dataset

Descriptive Statistics

Field	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
Count	1581.0	1581.0	1581.0	1581.0	1581.0	1581.0
Mean	31.922201	2.457938	60392.220114	20225.559322	79625.996205	35597.72296
Std	8.425978	0.943483	14674.825044	19573.149277	25545.857768	13633.63654
Min	22.0	0.0	30000.0	0.0	30000.0	18000.0
25%	25.0	2.0	51900.0	0.0	60500.0	25000.0
50%	29.0	2.0	59500.0	25600.0	78000.0	31000.0
75%	38.0	3.0	71800.0	38300.0	95900.0	47000.0
Max	54.0	4.0	99300.0	80500.0	171000.0	70000.0

Table 2 Descriptive Statistics

4) Data Cleaning

Missing Value Analysis and Treatment

The dataset was examined for missing values using the isnull( ) function. Missing values were found in 2 variables: Gender (53) and Partner_salary (106).

Standardization of inconsistent values:

It was found that the Gender variable contained inconsistent values. Examples of inconsistent values included variations in the same category, such as “Femal” and “Femle” for the category “Female”.
Such inconsistencies were handled by standardizing them to maintain uniformity.

Missing Value Treatment

Missing values in the Gender variable were treated by using the Mode (most frequently occurring category) method to preserve overall distribution.

The missing values found in the Partner_working column were handled using available information from related variables.

Partner_working	Count
No	90
Yes	16

Table 3 Missing values in Partner_working

The approach used for handling these missing values is as follows:

a) For records where Partner_working = ‘No’, the missing Partner_salary values must be entered as 0, since there is no means of income.
b) For the records where Partner_working = ‘Yes’ and Total_salary is greater than Salary, the missing Partner_salary can be derived by subtracting the individual’s salary from the total salary, i.e.,
Partner salary = Total salary – Individual salary

The approach makes sure that the missing values are filled with logically coherent information present in the dataset.

Following the treatment of missing and inconsistent data, the dataset was considered analysis-ready and suitable for further analysis.

Analysis

1. Univariate Analysis:

Univariate analysis was conducted to understand the distribution and characteristics of individual variables of the dataset.

Categorical Variables:

Frequency distribution was used for categorical variables.

- Make (Type of Automobile):

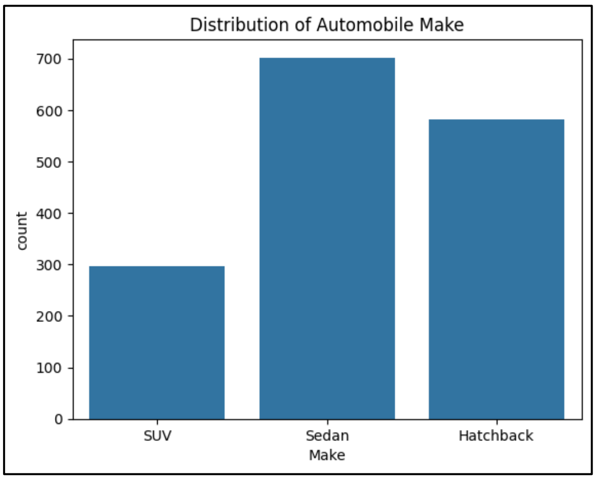

Figure 1 Distribution of Make

Sedan is the **most preferred automobile** make with the highest count, followed by Hatchback, while SUV records the lowest preference among the three categories.

- Gender

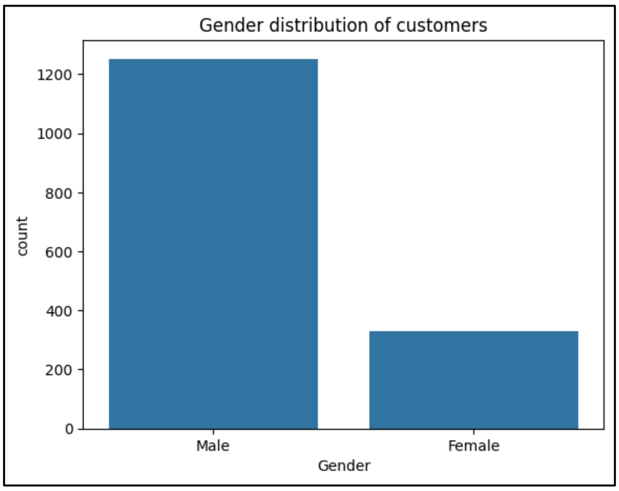

Figure 2 Gender distribution of Customers

Males dominate the automobile market, accounting for a significantly larger share of customers compared to females.

- Profession

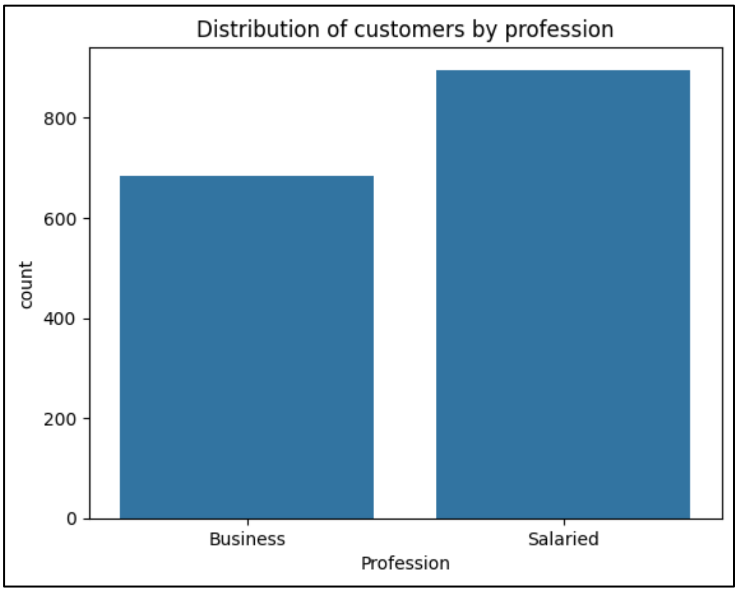

Figure 3 Distribution of customer profession

Salaried professionals represent the largest customer segment, outnumbering business professionals by a considerable margin.

- Marital Status

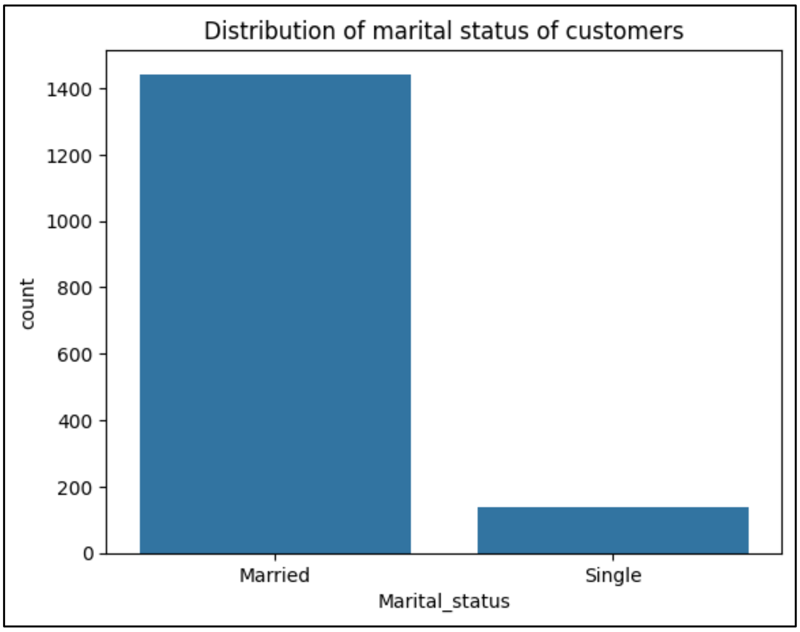

Figure 4 Distribution of marital status of customers

Married customers constitute the dominant segment of buyers, significantly outnumbering their single counterparts.

- Education

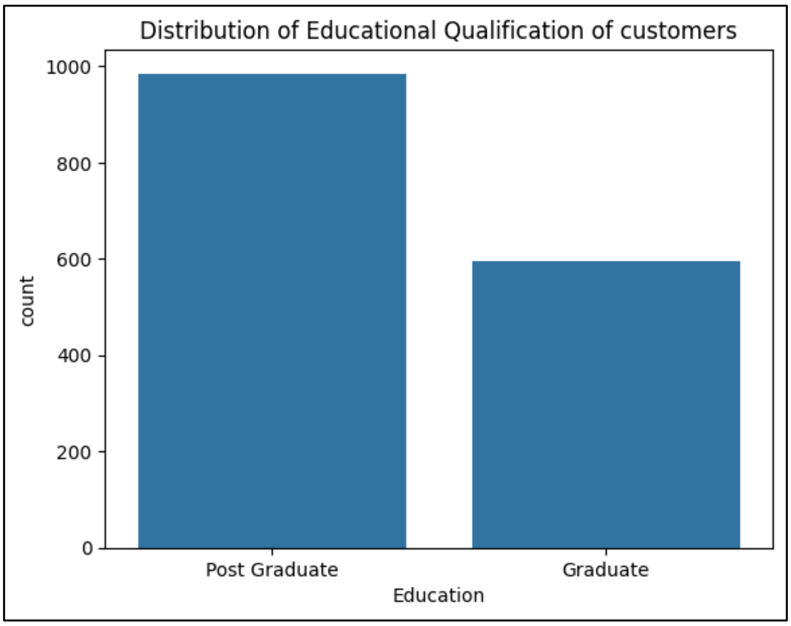

Figure 5 Distribution of Education qualification of customers

Postgraduates represent the largest and most dominant segment of automobile buyers.

- Personal loan

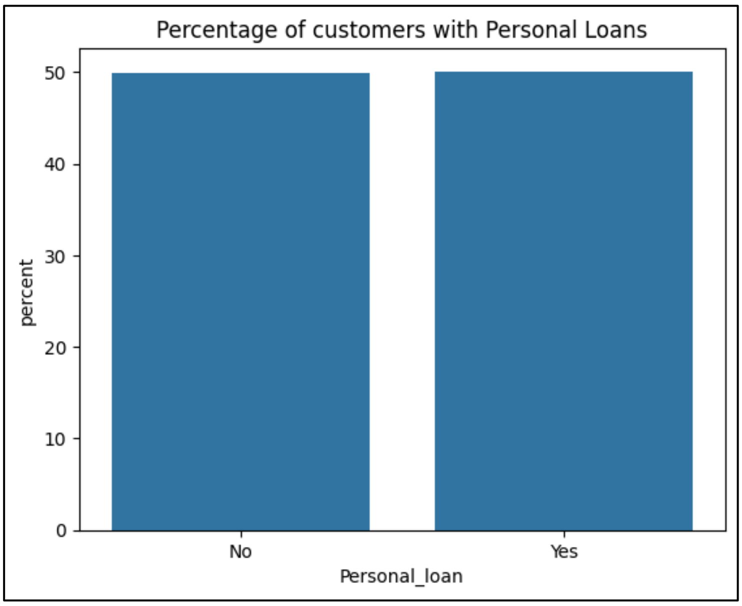

Figure 6 Percentage distribution of customers with personal loans

Data reveals **equal distribution** between customers who have taken personal loans and who have not, with each group accounting for approx. 50% of the total dataset.

- House loan

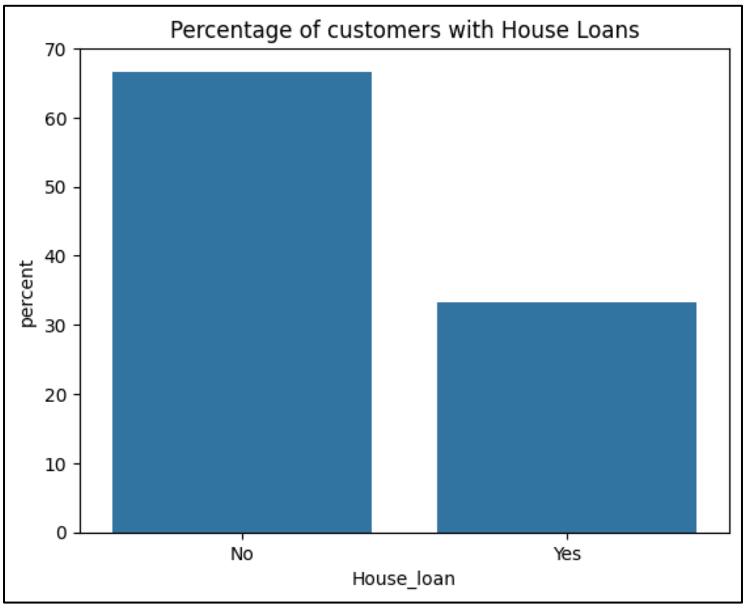

Figure 7 Percentage distribution of customers with house loans

The customers who have not taken house loans are significantly larger than those who have. Therefore, can conclude that most customers are unleveraged in terms of housing finance.

- Partner working

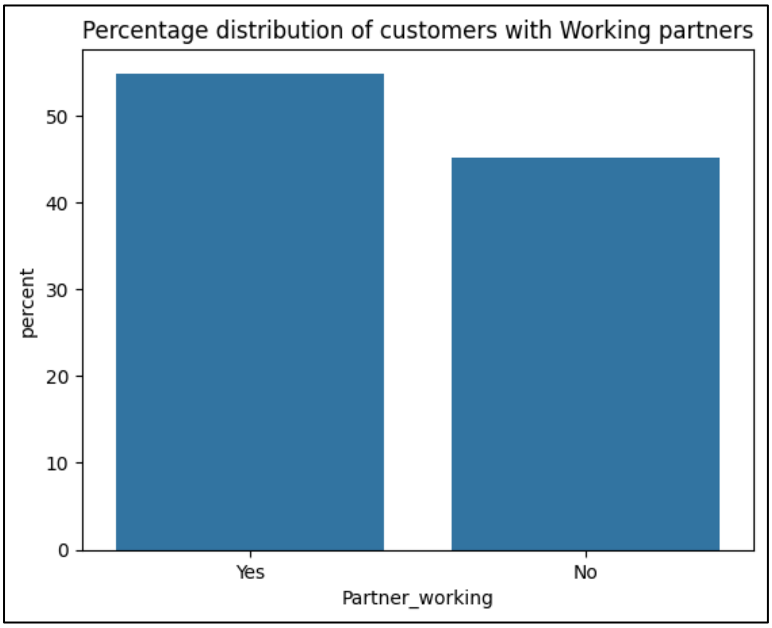

Figure 8 Percentage distribution of customers with working partners

Customers with working partners are in the majority, with a minor margin of nearly 4% difference to those who don't have a working partner.

Numerical Variables:

These variables were examined through statistical summaries and visualizations.

- Age

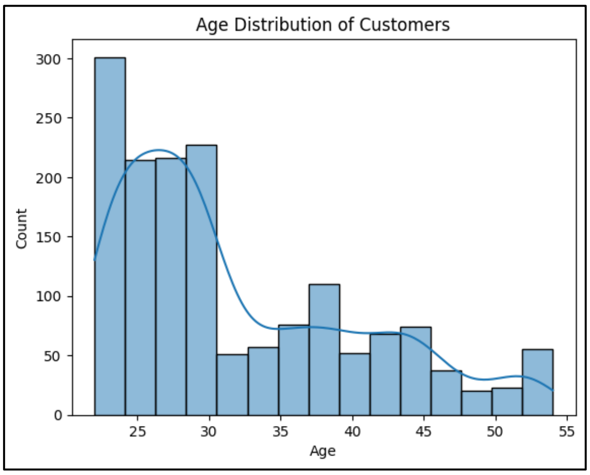

Figure 9 Age distribution of customers

The histogram indicates a right-skewed age distribution. The highest number of customers falls in the 25–30 age group, and the frequency gradually declines from the 35–55 range.

- Salary

Figure 10 Salary distribution of customers

Median customer salary is approximately 60,000, with most customers earning between 52,000 and 72,000. Salaries range from 30,000 to 100,000, indicating a diverse customer income base, and no significant outliers are observed.

- Partner Salary

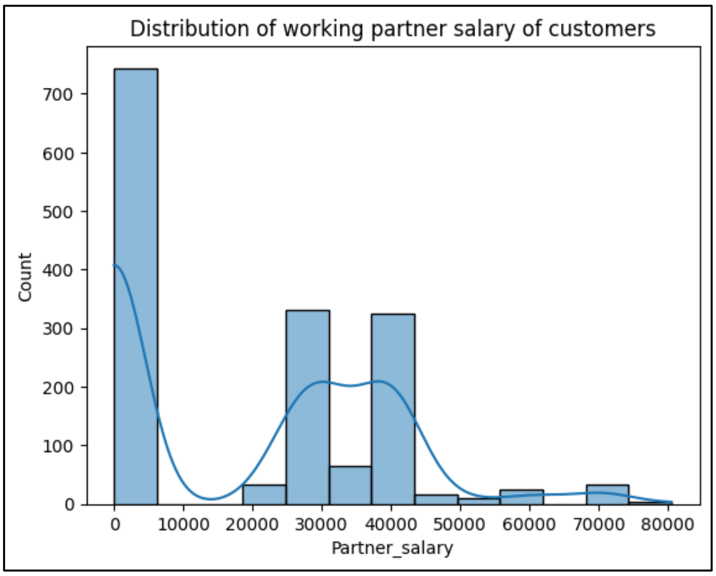

Figure 11 Distribution of working partner salary

The distribution is highly right-skewed, with a large concentration of values at the lower end, while a smaller group of customers has partner salaries ranging from 25,000 to 45,000 and a few high-income earners extending up to 80,000.

- Total Salary

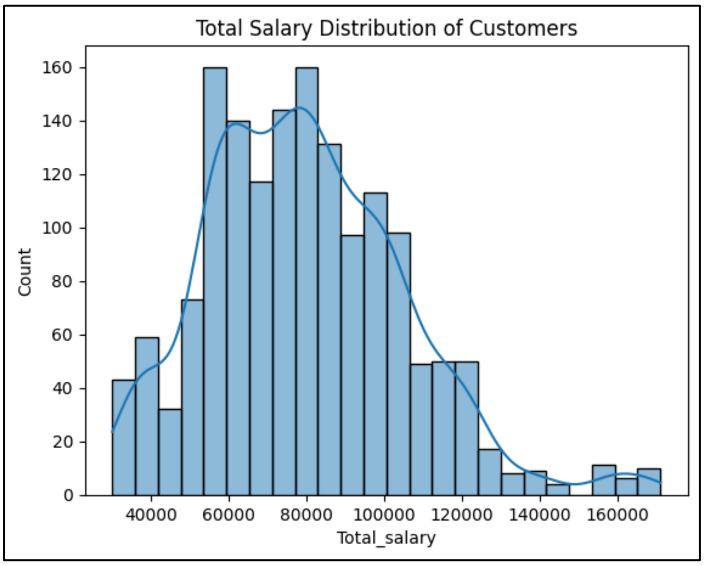

Figure 12 Total salary distribution of customers

The distribution of total salary is positively (right) skewed, with most customers having a total salary between 50,000 and 100,000, while a smaller chunk of customers earns substantially higher.

- Price

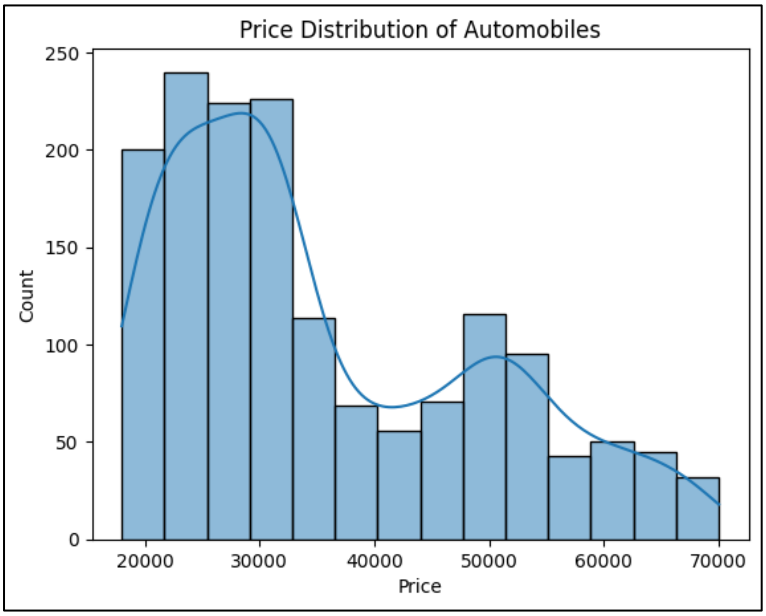

Figure 13 Price distribution of automobiles

The price distribution of automobiles is also positively skewed, with most of the models lying in the range of 20,000 and 35,000. The distribution extends to 70,000 by a small number of high-priced vehicles, and we can conclude that the lower- and mid ranged automobile attract customers.

2. Bivariate Analysis

Bivariate analysis was conducted to examine the relationship between the vehicle preference (Make) and the customer demographics and characteristics.

- Vehicle category and Gender

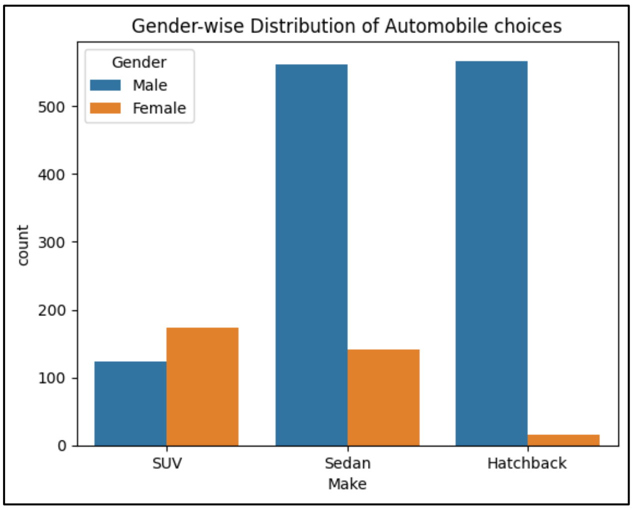

Figure 14 Gender-wise distribution of Make

The data reveals a clear gender-based distinction in automobile preferences. Males demonstrate a stronger inclination toward Sedans and Hatchbacks, whereas females show a significantly higher preference for SUVs.

- Vehicle category and Profession

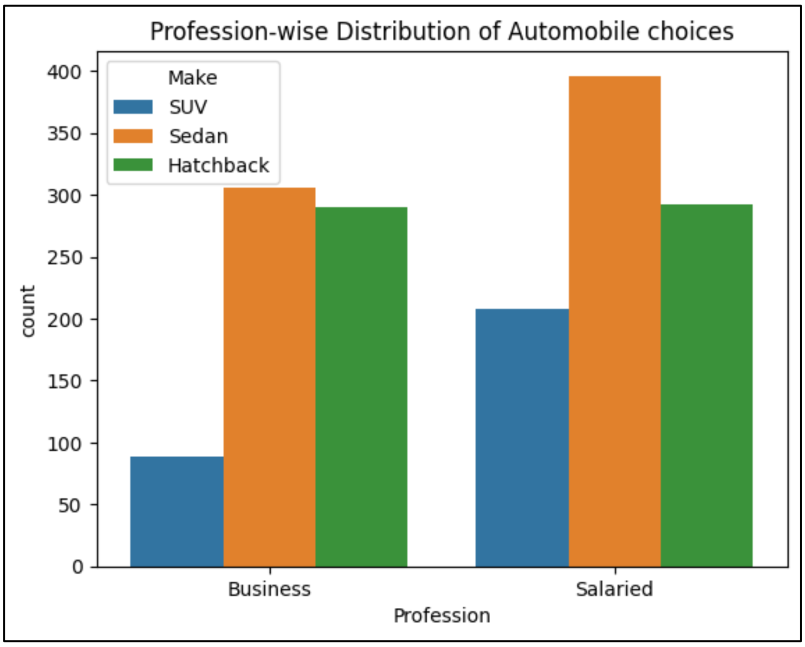

Figure 15 Profession-wise distribution of Make

Sedan is the most preferred automobile type across both professions, with the highest preference observed among **Salaried** individuals. This is followed by Hatchback and SUV, which rank second and third, respectively, in both the Business and Salaried categories.

- Vehicle category and Marital Status

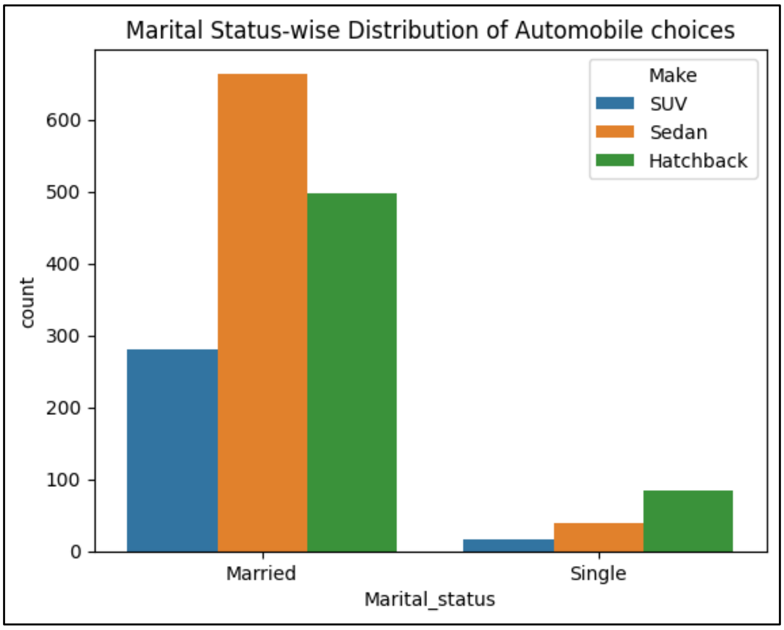

Figure 16 Marital status-wise distribution of Make

Sedans are highly preferred by married individuals, followed by Hatchbacks and then SUVs. In contrast, among Singles, Hatchbacks are the most preferred, with Sedans and SUVs following closely.

- Vehicle category and Number of Dependents

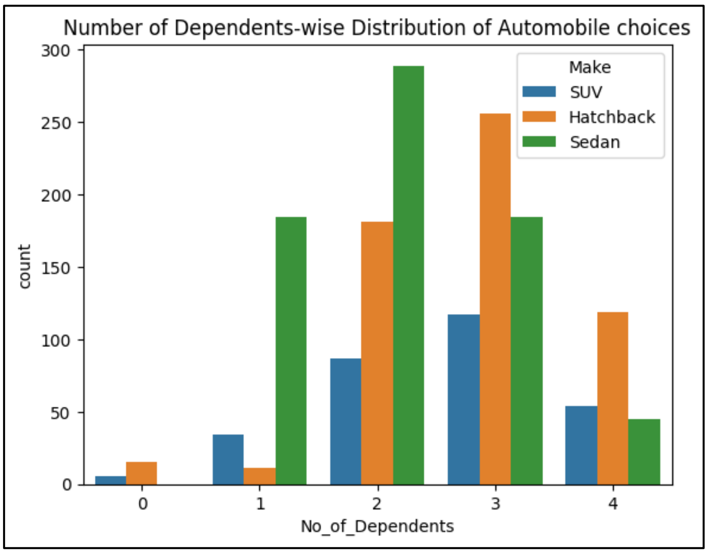

Figure 17 No of Dependents- wise distribution of Make

  - Customers with 2-3 dependents contribute to most of the automobile purchases in all the vehicle segments.
  - Customers with 1–2 dependents are most likely to purchase sedans, whereas customers with 3–4 dependents mostly buy hatchbacks.
  - All dependent groups have low SUV purchases, but are highest with customers with 3 dependents.

- Vehicle category and Personal loan

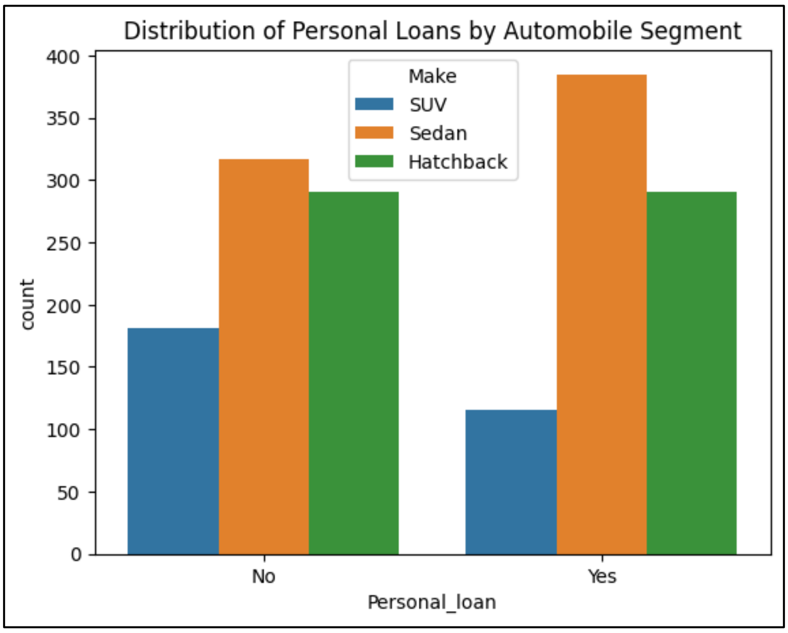

Figure 18 Distribution of personal loans by make

  - Sedans are the automobile segment that is purchased most often among customers with or without a personal loan.
  - Personal loan customers have a greater inclination toward buying sedans, whereas the purchases of Hatchbacks have been observed to remain steady whether a personal loan is taken or not.

- Vehicle category and Age

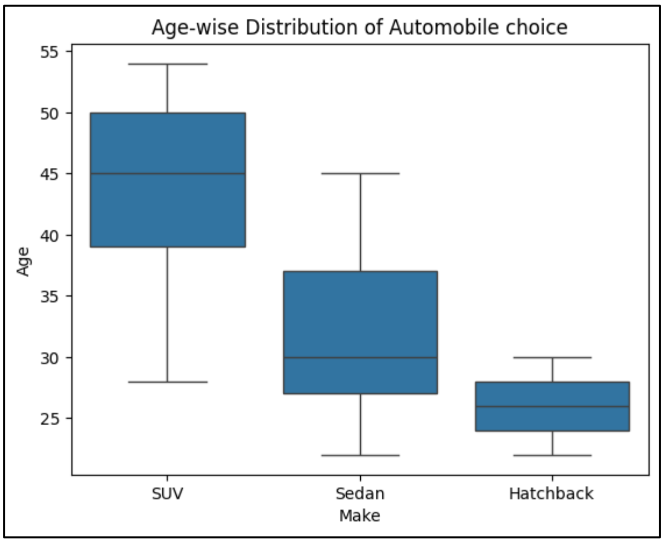

Figure 19 Age-wise distribution of Make

  - Customer age distribution varies significantly across vehicle segments.
  - SUV buyers exhibit the widest age range, spanning approximately 28 to 54 years, indicating broad appeal across different age groups.
  - Sedan customers are generally younger, with ages ranging from about 22 to 45 years.
  - Hatchback buyers show the narrowest age distribution, concentrated between 22 and 30 years, suggesting that this segment is most popular among younger customers.

- Vehicle category and Salary

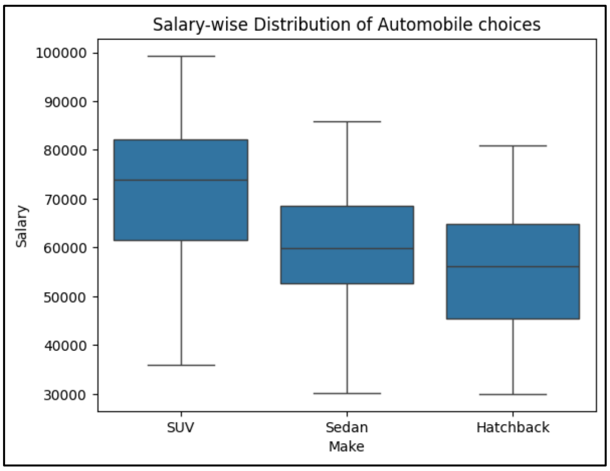

Figure 20 Salary-wise distribution of Make

  - SUV buyers demonstrate the highest salary range, with median salaries approximately between 65,000 and 80,000, suggesting a preference among higher-income customers.
  - Sedan customers exhibit a moderate salary range, with median values concentrated around 60,000 to 70,000.
  - Hatchback buyers display the lowest salary distribution, with median salaries ranging from approx. 55,000 to 65,000, indicating that this segment is predominantly favoured by budget-conscious customers.

- Vehicle category and Price

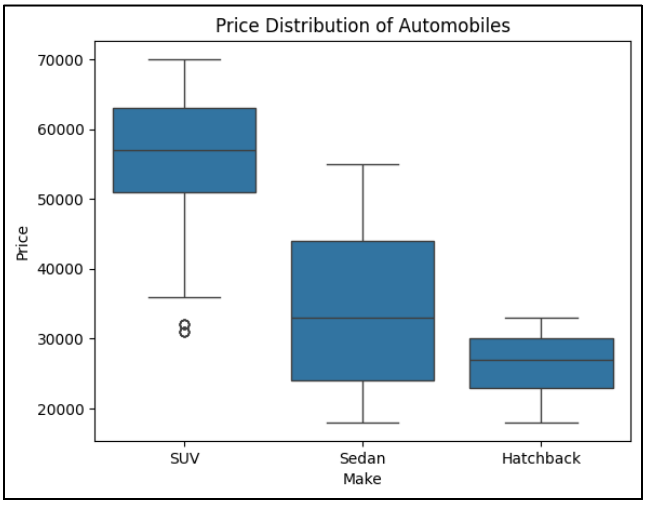

Figure 21 Price distribution of Make

  - SUVs are the highest-priced vehicle segment, with a median price of approximately 57,000 and a wider price range compared to other segments.
  - Sedans occupy the mid-price range, while Hatchbacks are the most affordable, exhibiting the lowest median price.
  - A few lower-priced SUV outliers are also observed.

- Vehicle category and Total Salary

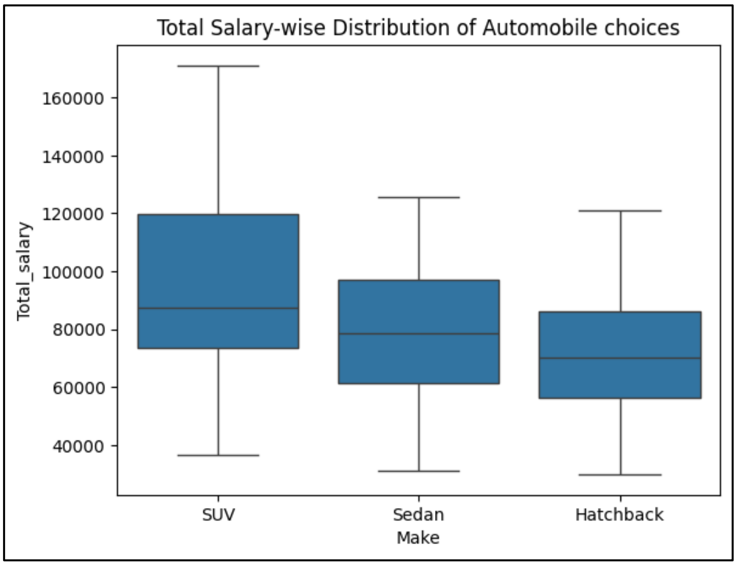

Figure 22 Total Salary-wise distribution of Make

  - Customers purchasing SUVs tend to have higher total salaries than buyers of other vehicle segments, with a median total salary of approx. 85000.
  - Sedan buyers fall in between, while Hatchback buyers generally have the lowest total salaries.

3. Multivariate Analysis

- Correlation Matrix

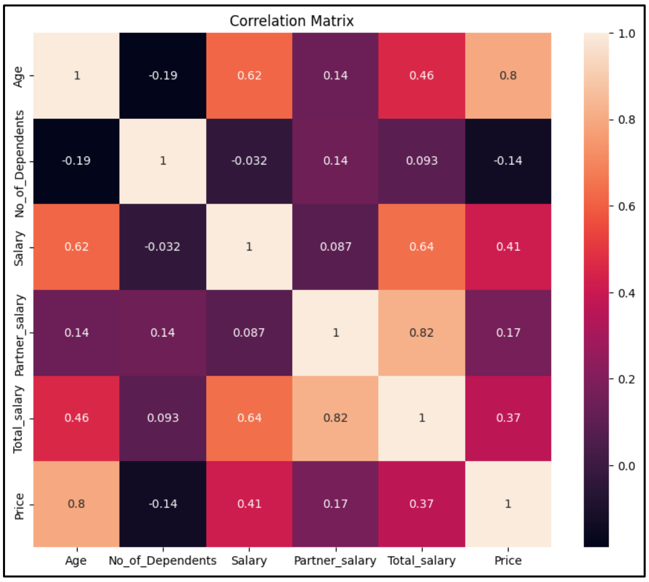

Figure 23 Correlation matrix

- Salary vs Price across Automobile segment

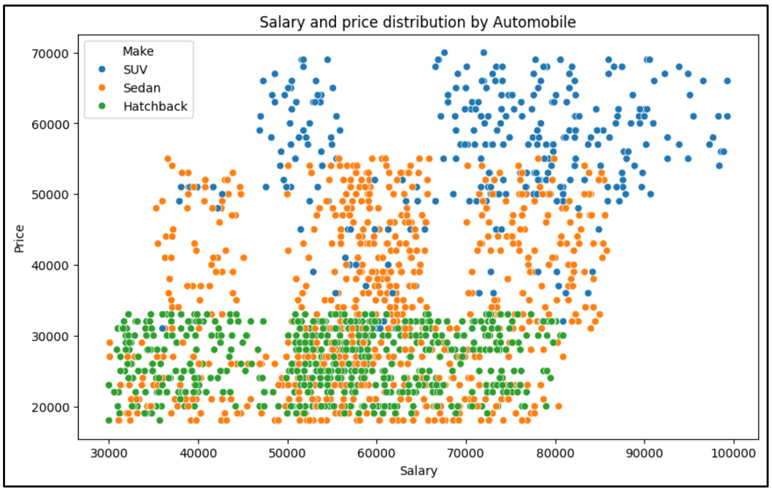

Figure 24 Salary and price distribution by Make

- Higher-income customers tend to opt for higher-priced automobile segments, with SUVs being the preferred choice.
- Hatchbacks are dense in the lower price range, while Sedans serve as the bridge in the middle-income segment.

Key Questions

1. Do men tend to prefer SUVs more compared to women?

→ Based on the Gender-wise distribution of Make (Figure 14), females show a stronger preference towards SUVs than males. While the total number of male buyers is higher, SUVs constitute a much larger proportion of purchases among female customers.

2. What is the likelihood of a salaried person buying a Sedan?

→ Based on the Profession-wise distribution of Make (Figure 15), Sedan is the most preferred automobile type by salaried professionals, which therefore indicates a higher chance of repeat purchases.

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for an SUV sale over a Sedan sale?

→

Make	Count
SUV	90
Sedan	305
Hatchback	277

Table 4 Salaried male count by Make

The above table was derived by the following logic: The dataset was first filtered for records that mapped to a Male who was salaried. Then the resulting subset was grouped by the Make of the automobile to calculate the count of male customers across each automobile segment.

Therefore, based on the result, Sheldon Cooper's claim that a male is an easier target for an SUV sale over a Sedan sale is false, as males prefer Sedan, followed by Hatchback and finally an SUV.

4. How does the amount spent on purchasing automobiles vary by gender?

→

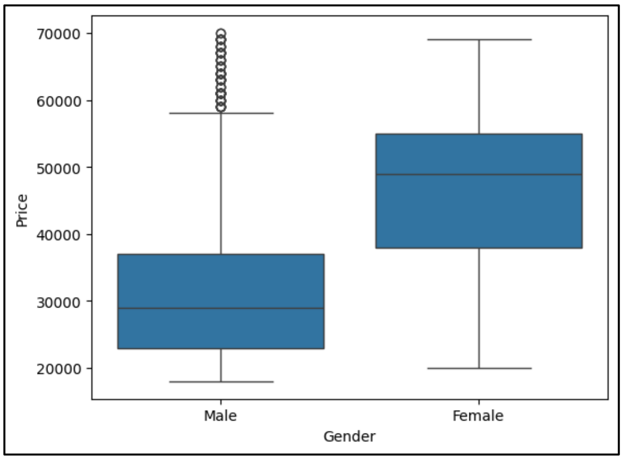

Figure 25 Automobile price distribution by Gender

From the above plot, the variation in automobile purchase by gender can be understood.
Since the median price of the female category is ~49,000 and the median price of the male category is ~29,000, we can conclude that female customers generally purchase higher-priced automobiles than male customers. Although there are a few outliers among male customers who are interested in higher-priced automobiles, the larger chunk of high-end purchases is by female customers.

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

→

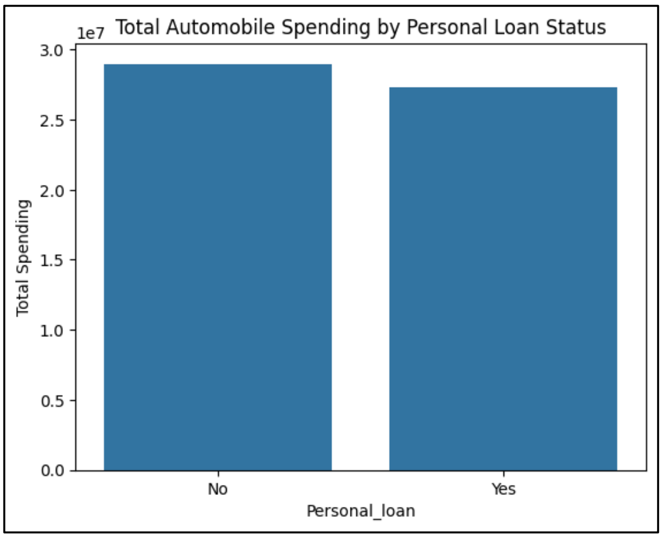

Figure 26 Spendings on Automobile by personal loan

From the above chart, the total amount spent on automobiles by customers who had taken a personal loan is approximately 2.7 million, and the total amount spent by customers who hadn't taken a personal loan is approximately 2.9 million. Although the difference is relatively small, both groups significantly contribute to automobile sales.

6. How does having a working partner influence the purchase of higher-priced cars?

→

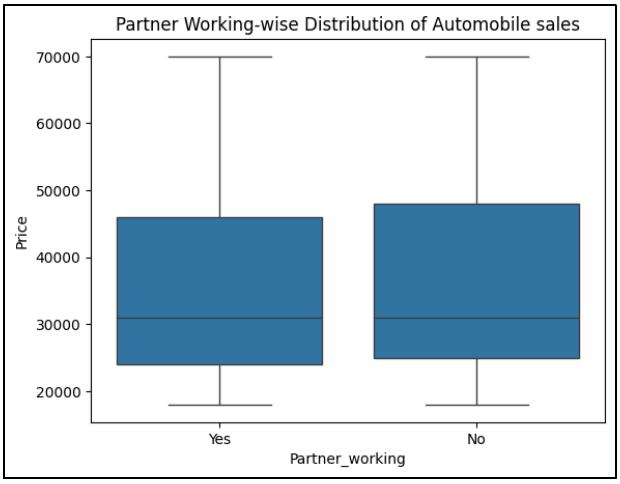

Figure 27 Partner working-wise distribution of automobile sales

From the above plot, it is seen that there is minimal variation between the 2 groups. The median purchasing price and overall price distribution are nearly identical across both groups who have a working partner and those who do not. Based on this, there is no strong influence of a working partner on the price of purchasing an automobile.

Key Actionable Insights

1) Sedan is the most preferred automobile category.
   - Across both business and salaried professionals, Sedan purchases are higher than SUVs and Hatchbacks, therefore indicating a strong customer base for the Sedan category.

2) Female customers show a higher preference for the SUV category.
   - Although male customers form the dominant customer base, a very high proportion of female customers opted for the SUV automobile than male.

3) Salaried customers are most likely to purchase Sedans.
   - Sedans have the highest share of purchases among salaried customers, making them the most preferred type in this customer segment.

4) Demand for mid-range and affordable vehicles.
   - Most automobile purchases occurred below 50,000, driven by Sedan and Hatchback sales, which highlights the strong customer demand for mid-range vehicles.

5) Sales of SUV and Hatchback categories.
   - Customers with 2 to 4 dependents account for the majority of purchases of SUVs and Hatchbacks, indicating that family size does influence preferences.

Business Recommendations

1. To strengthen the marketing effort for Sedan models.
   - Since Sedans represent the most preferred category, Austo should continue prioritizing Sedan-focused marketing campaigns.

2. Develop targeted campaigns for female SUV buyers.
   - Female customers show a relatively stronger preference for SUVs; marketing campaigns involving safety, comfort, and technology may improve engagement in this segment.

3. Focus on salaried professionals as a core customer segment.
   - A large portion of customers are represented by salaried professionals, and also show a strong inclination towards Sedan sales. Tailoring packages may increase conversion rates for this segment.

4. Focus on the high-demand mid-range segment.
   - Strengthen marketing efforts for Sedan and Hatchback models in the sub-50,000 segment, as it demonstrates the highest concentration of customer demand.

5. Validate marketing assumptions before implementation.
   - Targeting households with multiple dependents through family-centred campaigns that focus on space, comfort, and practicality in Sedan and Hatchback models can increase sales.